Gas Physics

What temperature and molecular weight do to exhaust speed, why a nozzle turns heat into motion, and why hydrogen wins.

Isagi · November 2026 · Theory from NASA Glenn's Beginner's Guide to Rockets; ISRO figures checked 9 November 2026

HOW TO USE THIS CHAPTER

- **Print single-sided** and write in the margins.
- Do each example with a calculator first. Temperatures are in kelvin (K = °C + 273.15).
- ISRO does not publish chamber temperatures, exhaust composition or nozzle pressure ratios. The values below are **illustrative, textbook-typical** numbers for each propellant type, chosen to check the physics against the Isp values from Chapter 1.

By the end of this chapter you can: use the ideal-gas law; explain why light gases move faster; derive and use the exhaust-velocity formula; and explain why a LOX/LH_2 engine reaches higher exhaust speed than a solid.

1 The gas law and what temperature means

NASA Glenn's thermodynamics pages start from the **equation of state**, which links pressure, density and temperature of a gas. In its form for a single gas:

$$p = \rho \times R_s \times T$$

$$R_s = R_u / M \qquad R_u = 8.314\ \text{J/(mol·K)} \qquad M = \text{molar mass in kg/mol}$$

p is pressure, ρ is density, T is absolute temperature, and R_s is the **specific gas constant**, which depends on the gas. For air, R_s = 287 J/(kg·K). Temperature, at heart, is a measure of how fast the molecules move: the average kinetic energy of a molecule is proportional to T. At the same temperature every molecule has the same average energy, so a **lighter molecule must move faster**. Its average speed goes as 1/√M.

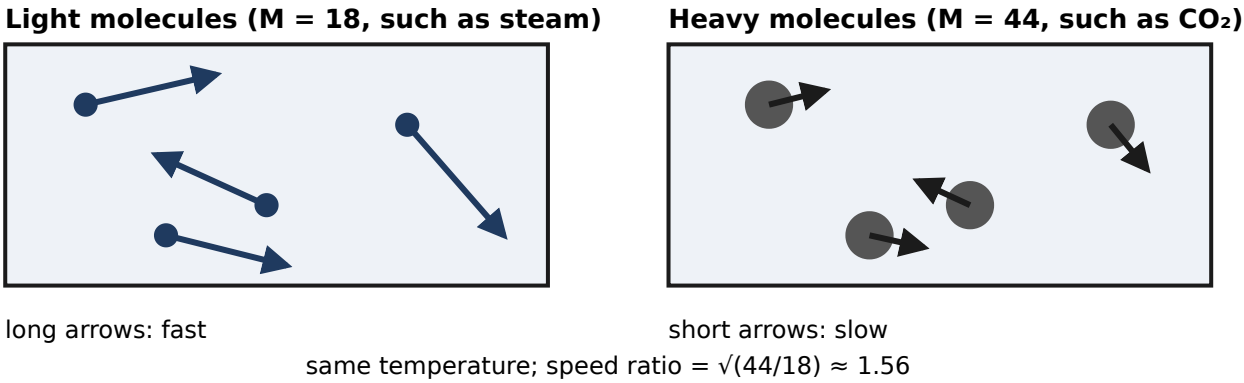


Figure 1. Same temperature, same average energy per molecule. Lighter molecules move faster, so a hot exhaust of light molecules gives higher exhaust speed.

2 Where the exhaust speed comes from

The first law of thermodynamics says energy is conserved: heat energy, work and kinetic energy can be exchanged but not created. Inside the chamber the gas is hot and almost still. As it flows through the nozzle it cools and speeds up. If the flow loses no heat to the walls and has no friction (an **isentropic** flow), the heat the gas gives up becomes kinetic energy of the stream. Per kilogram:

$$\tfrac{1}{2} v_e^2 = c_p \times (T_c - T_e)$$
$$c_p = \gamma/(\gamma - 1) \times R_s \quad \text{(specific heat at constant pressure)}$$
$$T_e/T_c = (p_e/p_c)^{(\gamma-1)/\gamma} \quad \text{(isentropic flow)}$$

Put them together and you have the standard exhaust-velocity formula:

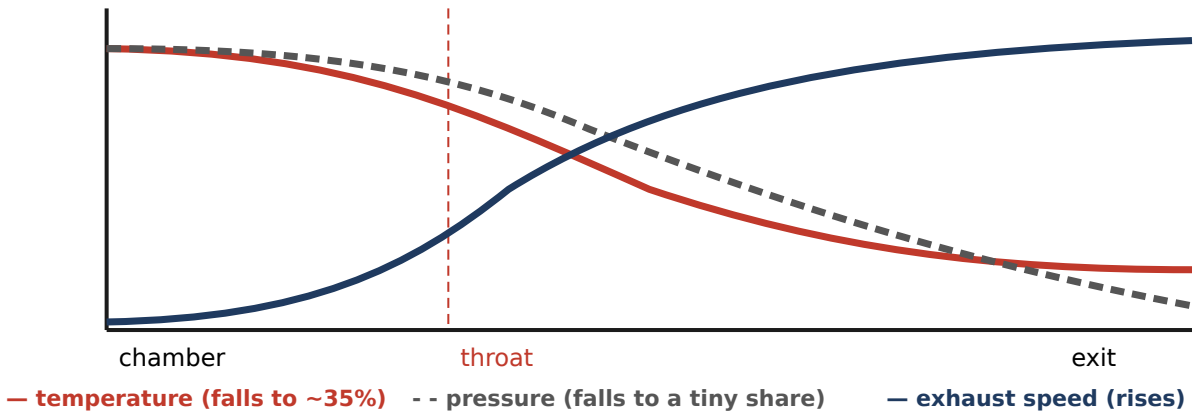
$$v_e = \sqrt{[\ 2\gamma/(\gamma - 1) \times (R_u/M) \times T_c \times (1 - (p_e/p_c)^{(\gamma-1)/\gamma})\ ]}$$

T_c, p_c = chamber temperature and pressure
p_e = pressure at the nozzle exit γ = ratio of specific heats (about 1.2)

Read it as a recipe. A **hotter chamber** raises v_e (as $\sqrt{T_c}$). A **lighter exhaust**, smaller M, raises it (as $1/\sqrt{M}$). A **bigger pressure drop**, a lower p_e/p_c, raises it by letting the nozzle pull out more of the heat. That last term is the nozzle's whole job (Chapter 1).

ISRO LENS: WHY THE C25 IS COLD AND FAST

A LOX/LH_2 flame is not hotter than a solid's. Both burn around 3,000 to 3,500 K. The C25 wins because its exhaust is mostly light steam and leftover hydrogen, with a low M, while a solid's exhaust holds heavier molecules and some solid particles. The $\sqrt{(T_c/M)}$ in the formula does the rest. Chapter 1's 443 s against 274.5 s is the result.



schematic, normalised; illustrative LOX/LH_2 values from Example C

Figure 2. Along the nozzle, temperature and pressure fall while speed climbs. The heat the gas gives up becomes motion.

THE SECOND LAW, IN A PARAGRAPH

NASA's thermodynamics pages also cover entropy and the second law, which says heat cannot be turned into work with 100% efficiency and that real flows always lose some energy to friction and heat. For rockets this means the isentropic formula is the ceiling, and real exhaust speeds come out a bit lower. The sums in Example C show how close the ceiling is.

3 Worked examples

Example A: the gas law and air density

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Sea level: ρ = 101,325 / (287 × 288.15) ≈ 1.225 kg/m³
At 11 km:  ρ =  22,632 / (287 × 216.65) ≈ 0.364 kg/m³
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These are the densities used in Chapters 3 and 4. Air at 11 km is cold and thin: less than a third of sea-level density.

Example B: light versus heavy at the same temperature

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Speed ratio = √(M_heavy / M_light)
Steam (18) against CO₂ (44):  √(44/18) ≈ 1.56
Hydrogen (2) against nitrogen (28):  √(28/2) ≈ 3.7
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Hydrogen molecules move almost four times faster than nitrogen at the same temperature. That is the underlying reason for LH_2's appeal.

Example C: compute v_e for a LOX/LH_2 stage and a solid

Illustrative values: LOX/LH_2 with T_c = 3,500 K, M = 13.5 g/mol, γ = 1.20, p_e/p_c = 0.002 (a vacuum nozzle). Aluminised solid with T_c = 3,400 K, M = 28 g/mol, γ = 1.18, p_e/p_c = 0.01.

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LOX/LH₂:  R_s = 8.314/0.0135 = 616 J/(kg·K)
   pressure term: 1 − 0.002^0.167 = 1 − 0.355 = 0.645
   v_e = √[12 × 616 × 3,500 × 0.645] ≈ 4,085 m/s
Solid:    R_s = 8.314/0.028 = 297 J/(kg·K)
   pressure term: 1 − 0.01^0.1525 = 1 − 0.495 = 0.505
   v_e = √[13.1 × 297 × 3,400 × 0.505] ≈ 2,585 m/s
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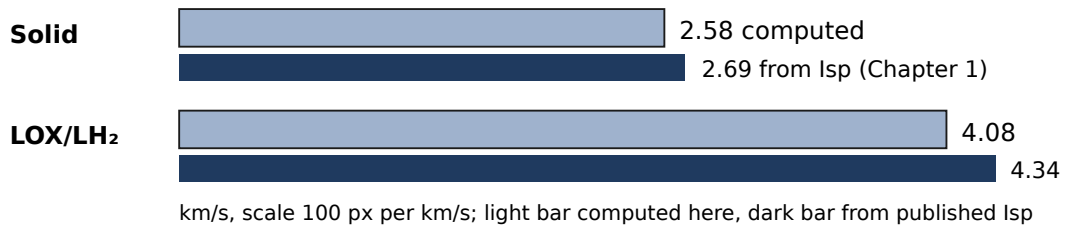


Figure 3. The simple formula lands within about 5% of the exhaust speeds implied by the stage Isp values.

The ratio of the two computed speeds is 4,085 / 2,585 ≈ 1.58, close to the 443 / 274.5 = 1.61 from Chapter 1. The model is crude (real exhaust composition and temperature differ), but it reproduces the gap and shows where it comes from: mostly M, with a little from T_c and the pressure term. The computed values are slightly below the Isp-based ones, so the assumed M, T_c and γ are only rough stand-ins.

Example D: how much of the heat becomes motion?

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LOX/LH₂:  T_e = 3,500 × 0.355 ≈ 1,242 K
          heat turned into motion = 0.645 ≈ 65%
Solid:    T_e = 3,400 × 0.495 ≈ 1,684 K
          heat turned into motion = 0.505 ≈ 50%
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A bigger expansion, to a much lower exit pressure, converts about 65% of the heat in the LOX/LH_2 case, against about 50% in the solid case, and leaves the exhaust much colder. The rest leaves as hot gas.

CHAPTER SUMMARY

- Gas law: $p = \rho R_s T$, with $R_s = R_u/M$. Temperature measures molecular motion; lighter molecules move faster at the same T (speed $\propto 1/\sqrt{M}$).
- Energy conservation in an isentropic nozzle gives $v_e = \sqrt{[2\gamma/(\gamma-1) \times (R_u/M) \times T_c \times (1 - (p_e/p_c)^{(\gamma-1)/\gamma})]}$.
- v_e rises with chamber temperature ($\sqrt{T_c}$), falls with molar mass ($1/\sqrt{M}$), and rises with a bigger pressure drop.
- LOX/LH_2 is not hotter than a solid; its exhaust is lighter. That is why the C25 reaches 443 s.
- Illustrative sums reproduce the 1.6 : 1 exhaust-speed gap between C25 and S200 to within a few percent.

Next, Chapter 7: the capstone. SSLV, PSLV and LVM3 compared with everything from Chapters 0 to 6.

Practice problems

R_u = 8.314 J/(mol·K). Answers are on the last page.

P1. Air at 300 K and 101,325 Pa (R_s = 287). What is its density?

P2. Speed of sound $a = \sqrt{(\gamma R_s T)}$ at 300 K for air (γ = 1.4, R_s = 287) and for hydrogen (γ = 1.4, R_s = 4,124).

P3. In Example C's LOX/LH_2 case, lower T_c to 3,000 K. What is v_e, and what is the ratio to the 3,500 K case?

P4. Keep T_c = 3,500 K but let the exit pressure ratio be 0.005 (a shorter nozzle). Find T_e and v_e.

P5. If a new fuel gave an exhaust of M = 7.5 g/mol instead of 13.5 at the same T_c and pressure ratio, by what factor would v_e change?

P6. Why does the formula say the nozzle exit pressure matters, and when is more expansion no longer useful?

Glossary

Equation of state
$p = \rho R_s T$, linking pressure, density and temperature.

Specific gas constant (R_s)
R_u/M; depends on the gas.

Molar mass (M)
Mass of one mole of the gas, in kg/mol or g/mol.

Specific heat ratio (γ)
c_p/c_v, about 1.2 for rocket exhaust and 1.4 for air.

Isentropic flow
Flow with no heat loss and no friction.

Chamber conditions (T_c, p_c)
Temperature and pressure in the combustion chamber.

Exit pressure (p_e)
Pressure of the gas leaving the nozzle.

Expansion
The gas widening in the bell as its pressure falls.

Entropy
A measure of energy spread out and unavailable; always rises in real flows.

Kelvin
Absolute temperature scale; 0 K = −273.15 °C.

Answer key

Fold this page back before you start the problems.

P1. $\rho = 101{,}325 / (287 \times 300) \approx 1.18$ kg/m³.

P2. Air: $\sqrt{(1.4 \times 287 \times 300)} \approx 347$ m/s. Hydrogen: $\sqrt{(1.4 \times 4{,}124 \times 300)} \approx 1{,}316$ m/s, about 3.8 times faster.

P3. $v_e \approx 3{,}782$ m/s. Ratio to 4,085 = 0.926; this is $\sqrt{(3{,}000/3{,}500)} = 0.926$, because $v_e \propto \sqrt{T_c}$.

P4. $T_e = 3{,}500 \times 0.005^{0.167} \approx 1{,}447$ K. $v_e \approx 3{,}895$ m/s, about 4.6% lower than the longer nozzle's 4,085 m/s.

P5. $\sqrt{(13.5/7.5)} \approx 1.34$, so v_e would be about 34% higher.

P6. A lower exit pressure means the nozzle extracts more of the gas's heat as motion, so v_e is higher. But the gain shrinks as p_e/p_c falls, and a larger bell adds weight and, at sea level, can let outside air push in. So nozzles are sized to the altitude they work at (Chapter 1).

Sources

Theory: NASA Glenn Research Center, Beginner's Guide to Rockets (equation of state, specific heats, first and second laws, isentropic flow, nozzle performance). Exhaust speeds implied by stage I_{sp} are from Chapter 1 (public references). Chamber temperatures, molar masses, γ and pressure ratios used here are illustrative textbook-typical values, not ISRO data; ISRO does not publish them.